

DELETE vs TRUNCATE vs DROP

BYTE BATTLE / QUIZ PREP

DBMS Revision Notes & Exam Reference Sheet

1. Basic Classification ★★★

Command	Category	Main Purpose
DELETE	DML (Data Manipulation)	Deletes specific or all rows/tuples from a table
TRUNCATE	DDL (Data Definition)	Removes all rows at a table level (fast)
DROP	DDL (Data Definition)	Removes the entire table object and schema completely

🧠 Easy Memory Trick:

- **DELETE** → Rows selectively (using `WHERE`)
- **TRUNCATE** → All rows instantly (Structure remains)
- **DROP** → Entire table disappears (Structure + Data gone)

2. Command Breakdown & Syntax

DELETE (DML)

Removes row(s) from a table.

```
DELETE FROM
Student;
```

Deletes all rows. Table structure remains intact. `DESC Student;` still works.

```
DELETE FROM
Student
WHERE ID = 1;
```

★ Supports `WHERE` clause for selective row deletion.

TRUNCATE (DDL)

Removes **all rows** from a table.

```
TRUNCATE TABLE
Student;
```

Rows → Gone ✗
Structure → Remains ✔ (Can still `DESC Student;`)

✗ **Does NOT support `WHERE` condition!**
`TRUNCATE TABLE Student`
`WHERE ID = 1;` (Syntax Error)

DROP (DDL)

Removes **entire table object**.

```
DROP TABLE
Student;
```

Table no longer exists! Structure, data, and column definitions are permanently removed.

Trying `DESC Student;` will result in a `Table does not exist` error.

3. Table State After Execution ★★★

Initial Table (Student)

ID	Name
1	A
2	B
3	C

After DELETE FROM Student;

ID	Name
(empty - 0 rows)	

✔ Structure Remains

After TRUNCATE TABLE Student;

ID	Name
(empty - 0 rows)	

✔ Structure Remains

After DROP TABLE Student;

✗ Table Does Not Exist

✗ Structure & Data Gone

4. DELETE vs TRUNCATE (Key Nuances)

Feature	DELETE	TRUNCATE
Type	DML	DDL
WHERE Clause	✔ Supported	✗ Not Supported

5. DROP vs TRUNCATE (Exam Favorite) ★★★

Feature	TRUNCATE	DROP
Category	DDL	DDL
Deletes Rows	✔ Yes	✔ Yes